



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

A.Y. 2026 – 2027

B.Tech. III Year Odd Semester

COURSE: CNAD (24CND3101)

PROJECT ABSTRACT SUBMISSION

TITLE OF THE PROJECT	CloudChange: A Cloud-Native Framework for Dependency-Aware Change Impact Analysis and Safe Kubernetes Deployment	
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Abstract

Cloud-native applications are built using multiple microservices, containers, databases, and Kubernetes resources that depend on each other. A small change in one component can therefore affect other services and cause deployment failures, configuration errors, or service downtime. CloudChange is proposed as a cloud-native framework for analyzing the impact of changes before they are deployed. The main objective of the project is to identify affected components, understand their dependencies, and determine the risk associated with a proposed change. The system analyzes application code, configuration files, container images, and Kubernetes manifests using dependency analysis and a dependency graph. It calculates an impact and risk score and provides recommendations such as proceeding with deployment, performing additional testing, or blocking the change for manual approval. CloudChange can be integrated with CI/CD tools and Kubernetes to support automated validation, health checks, controlled deployment, and rollback mechanisms. The framework aims to improve deployment reliability, reduce unexpected failures and downtime, and provide developers and DevOps teams with better visibility into system dependencies. The expected outcome is a safer, more controlled, and efficient Kubernetes deployment process.

Signature of the Guide

Signature of the Course Coordinator

HOD

Tools and Technologies Used for the Project:

Category	Tools / Technologies
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Java Spring Boot, Python, FastAPI
Database	PostgreSQL
Version Control	Git, GitHub
Containerization	Docker
Container Orchestration	Kubernetes / Minikube
Package Management	Helm
CI/CD	GitHub Actions
Security Scanning	Trivy
GitOps	Argo CD
Cloud Platform	AWS / Microsoft Azure / Google Cloud
API Testing	Postman
Development IDE	Visual Studio Code / IntelliJ IDEA
Monitoring	Prometheus, Grafana
Logging	ELK Stack
Configuration Management	YAML, Kubernetes Manifests
Deployment Strategy	Rolling Update, Health Checks, Automated Rollback
Documentation	GitHub, Markdown

Signature of the Guide**Course Coordinator****HOD**